Business Intelligence – Lab 2

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Abstract:

Nota Bene: This document is designed as an autonomous capsule. It contains lecture parts – explaining concepts – and lab parts – practical applications through coding exercises.

⚠️ **Important remark**: This capsule will use a set of icons and graphical chart to indicate several levels of information.

- 👉: Something that you should **do** to complete the Lab
- 💾: You need to **validate** something (call the teacher, save something for deposit)
- 💡: Tips or **useful** remark to help you in your lab
- 🧠: An explanation on a new knowledge, that you should probably know/**retain**
- ⚠️: Something important that you **need** to read
- ⭐: Something to try if you want to **explore** further on the subject
- IT SHOULD NOT BE NEEDED TO BE WRITTEN BUT EVERYTHING ON WHITE BACKGROUND SHOULD BE READ ALSO! (Only ⭐ is optional).

Preamble: Project environment

Welcome to your **Business Intelligence capstone project**!
Throughout this module, you won't just *learn* BI — you will **build your own BI solution**, just like in a real Data team.
You will work **in pairs** and progress step by step during the lab sessions.

🎯 The mini-project concept

During the project, you will define a business need (your choice), select a dataset, and then build:

- A requirements document
- An agile backlog
- A complete BI pipeline (ingestion → transformation → model)
- A final dashboard

By the end, you will be able to lead a BI project from A to Z, just like in a company.

💡 Lab 2 – User Stories, Backlog, Mock-up

In this second lab, you will continue preparing your BI project like a real Data team.
You will:

- write **User Stories**
- build an **agile backlog**
- start **thinking about how your tables should be organized for BI** (conceptual only)
- create a **dashboard mock-up**

⚠️ **Important**
You have *not yet* learned data modelling (semantic model, star schema, transformations).
This will be taught in **CM2**.
Today, we only work at a **conceptual business level**.

The objective: **have your dataset ready to be transformed and modelled in Lab 3.**

Task 1: Write User Stories

User Stories are used to express a clear business need, in a standard format commonly used in companies.

👉: Take your word document from yesterday, add a section titled “User Stories” and write **3 to 5 mandatory User Stories,** covering at least 2 different user profiles.

🎯 Required format:

As a [user],
I want to [action],
so that [business objective].

💡 **Examples:**

- As an HR Manager, I want to visualize the number of hires per month, so that I can monitor the evolution of the candidate pipeline.

- As a Sales Director, I want to see revenue by segment, so that I can adapt my sales strategy.

👉 For each User Story, you must add **Acceptance criteria (3–5 bullet points)**

💡 **Example:**

- The KPI must update automatically

- Filters must work on geographic areas

- The chart must allow drill-down

⚠️ **Reminder**
You will deposit your file on Moodle at the end of the lab.
Make it clean and readable. Explain your reasoning and the steps of your work.

⚠️ : **Don't forget to cite any AI generated code**.

Task 2: Build the Agile Backlog

You will now break down your User Stories into **technical tasks**, exactly like in a real BI project.

👉 Add a section **"Agile Backlog"** to your Word file.

Break down each User Story into **future tasks** that your BI project will require.

🧠 The backlog should contain:

- User Stories (ref. Task 1)

- Related technical tasks (you can sort them by pipeline stage: ingestion, transformation, modelling, reporting)

💡 Use simple, intuitive BI vocabulary (you have not learned modelling yet 😊

Task 3: Conceptual modelling

This is **not modelling**, only a **reflection** to prepare and help you for the next courses.

👉 Answer the following questions:

1. What seems to be your main activity table?

🧠 The table that contains events, actions or measurements (sales, matches, orders, readings, etc.)

2. Which tables describe things?

People, products, categories, places, dates, etc.
(Do NOT use the word “dimension”, just describe intuitively.)

3. What information will need to be cleaned or harmonized?

Dates, categories, units, duplicates, missing values…

4. What information may need to be grouped or transformed?

Examples:

- extract year/month from dates

- group product categories

- unify city names

- derive age groups

5. Which tables feel connected? (intuitive only)

Explain things like:

“Table A contains information that describes elements from Table B.”

👉 Draw a first simple conceptual schema/model

Task 4: Create a Dashboard Mock-up

👉 Add a simple mock-up of your future dashboard.

💡 You can:

- draw it by hand,

- use PowerPoint,

- or even create a basic diagram in Word.

⚠️ **This is not a real report**: just a sketch to represent how you imagine the future BI dashboard.

The mock-up must show:
- the KPIs you plan to display,
- 2–3 possible charts,
- the overall structure (header, filters, visualization areas).

💡 ***Objective: start thinking about the user experience before even opening Power BI.***

⭐ Tip: Your mock-up does not need to be perfect — it just needs to help visualize your ideas.

💾 : Remember to upload your Word file on Moodle